

Notes:Fang, Lerner, and Wu (RFS, 2017)

Intellectual Property Rights Protection, Ownership, and Innovation: Evidence from China

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1 Paper at a glance: question, idea, and punchline

Main question. How does the strength of *local* intellectual property rights (IPR) protection affect firms’ incentives to innovate, and how does this interact with *ownership structure* (state-owned vs private)?

Core idea. Use China’s large wave of state-owned enterprise (SOE) privatizations as a quasi-exogenous shock to ownership incentives. Then ask whether the post-privatization change in innovation is larger in regions where IPR protection is stronger. This yields a difference-in-differences (DiD) logic:

$$\text{Privatization shock} \times \text{Local IPR quality} \Rightarrow \text{Innovation response.}$$

Main findings (high-level).

1. **Privatization is associated with higher innovation.** Firms’ patent stocks are substantially higher in the five years after privatization than in the five years before.
2. **Local IPR protection is a key moderator.** The post-privatization increase in patenting is concentrated in high-IPR regions; in low-IPR regions, privatization alone does not robustly raise innovation.
3. **Evidence survives stronger designs.** Results persist with: (i) rich controls and fixed effects, (ii) matched-sample analyses enforcing parallel pre-trends, and (iii) instrumental variables (IV) for IPR protection.

2 Data construction and key objects

2.1 Local IPR protection score (prefecture-year)

Data source. The Chinese Academy of Social Sciences (CASS) Urban Competitiveness Reports (2002–2011) provide a prefecture-year index of perceived IPR enforcement quality, based on a survey of legal professionals and corporate executives.

Survey questions. Respondents rate (from 5 best to 1 worst):

1. time to resolve IP disputes in courts,
2. cost of dispute resolution relative to IP value,
3. fairness of court decisions.

Step 1: convert the distribution of responses into a single “prefecture response”.

Let $w_{p,q,t}^{(k)}$ be the share of respondents in prefecture p (question q , year t) choosing response $k \in$

$\{1, 2, 3, 4, 5\}$. The report maps the 1–5 scale into a centered score in $\{-2, -1, 0, 1, 2\}$:

$$\text{PrefectureResponse}_{p,q,t} = (-2)w_{p,q,t}^{(1)} + (-1)w_{p,q,t}^{(2)} + 0 \cdot w_{p,q,t}^{(3)} + 1 \cdot w_{p,q,t}^{(4)} + 2 \cdot w_{p,q,t}^{(5)}. \quad (1)$$

Step 2: average across the three questions.

$$\text{AvgResponse}_{p,t} = \frac{1}{3} \sum_{q=1}^3 \text{PrefectureResponse}_{p,q,t}.$$

Step 3: rescale to a unit interval by dividing by the sample maximum.

$$\text{IPRScore}_{p,t} = \frac{\text{AvgResponse}_{p,t}}{\max_{p' \in \mathcal{P}} \text{AvgResponse}_{p',t}}, \quad (2)$$

where the denominator is the maximum over all prefectures in the survey sample (66 prefectures). Hence $\text{IPRScore}_{p,t} \in [0, 1]$ and can be read as “percent of the best-practice prefecture” in a given year.

Reading Figures 1–2. Figures 1–2 visualize substantial geographic and time-series variation in $\text{IPRScore}_{p,t}$, which is crucial for identification.

2.2 Innovation: patent flows and patent stocks

Data source. Firm-level patent applications and grants from the Chinese State Intellectual Property Office (CSIPO).

Patent types used.

- **Invention patents:** higher novelty content; longer examination (18–36 months); lower success rate.
- **Utility patents:** incremental innovation / applications of existing tech; faster examination (3–6 months); higher success rate.

Why a stock measure? Innovation has persistence and patents are long-lived intangible assets. Following the standard perpetual-inventory logic (Griliches/Hall), construct a stock with depreciation.

Let $K_{i,t}$ denote firm i ’s patent stock in year t and let $n_{i,t}$ be the count of *ultimately granted* patents applied for in year t . Then:

$$K_{i,t} = (1 - \delta)K_{i,t-1} + n_{i,t}, \quad \delta = 0.15. \quad (3)$$

They construct stocks for: (i) invention-only, (ii) utility-only, and (iii) invention+utility.

2.3 SOE privatizations: ownership shocks

Data sources.

- **Listed firms:** CSMAR/WIND (CRSP/Compustat-style).
- **Unlisted industrial firms:** Chinese Industrial Enterprises Database (National Bureau of Statistics), matched across years using IDs when available and otherwise exact matches on name, sector, province, city, address.

Operational privatization definition (unlisted). A firm is an SOE if state equity share $> 50\%$. A privatization occurs when next year state share falls below 50% and another ownership category becomes dominant (or, when detailed shares are missing, “registration type” changes from SOE to private).

Event window. For each privatization, the analysis uses five years pre and five years post.

3 Economic mechanism and identification logic

3.1 Mechanism in one sentence

Privatization increases the residual-claim and control rights of private owners, which strengthens incentives to undertake (costly) innovation—*but only if* innovators can appropriate returns, which requires sufficiently strong local IPR protection.

3.2 Why privatization helps identification

A cross-sectional comparison of private vs SOEs is confounded because:

- private firms may sort into better regions,
- private firms may differ systematically in unobserved innovation capacity,
- regions may improve IPR in response to local innovation.

SOE privatizations create within-firm changes in ownership structure that are plausibly less tied to contemporaneous innovation motives (supported by the timing evidence in Figure 4).

4 Empirical models in the paper (all main equations)

4.1 Model 1: baseline DiD regression (paper Eq. (4))

Restrict to firms that experience a privatization in 2002–2011, keep years $t \in [-5, +5]$ around the event, and estimate:

$$\begin{aligned} \log(K_{i,t} + 1) = & \alpha + \beta_1 \text{POST}_{i,t} + \beta_2 \text{IPRScore}_{p(i),t-1} \\ & + \beta_3 (\text{POST}_{i,t} \times \text{IPRScore}_{p(i),t-1}) + \Gamma' X_{i,t-1} + \Lambda' Z_{p(i),t-1} + \mu_{\text{industry}} + \mu_{\text{province}} + \varepsilon_{i,t}. \end{aligned} \quad (4)$$

Key variables.

- $\text{POST}_{i,t} = 1$ for post-privatization years, 0 otherwise.
- $p(i)$ is firm i 's prefecture.
- $X_{i,t-1}$ includes (lagged) $\log(\text{assets})$, intangible/asset, ROA, leverage, age.
- $Z_{p,t-1}$ includes provincial GDP growth, university density, and government R&D subsidy growth.

Interpretation of the coefficients.

- β_1 is the post-privatization change in innovation when $\text{IPRScore} = 0$ (a baseline intercept; not meant to be literally interpreted at 0, but useful for the interaction).
- β_3 is the *moderation* effect: how much stronger the post-privatization effect is when local IPR protection is higher.
- The marginal post-privatization effect at IPR level s is:

$$\frac{\partial \log(K_{i,t} + 1)}{\partial \text{POST}_{i,t}} = \beta_1 + \beta_3 s.$$

Inference. Standard errors are clustered by province; tables report p -values in parentheses.

4.2 Model 2: matched-sample “dynamic DiD” (paper Eq. (5))

Motivation: even among privatized firms, high-IPR regions might systematically host firms with stronger *preexisting* innovation trends. The paper therefore matches each “treatment” privatization (in a high-IPR prefecture) to a “control” privatization (in a low-IPR prefecture) in the *same year* using propensity score matching on pre-privatization characteristics (including pre-trend in patent growth).

After matching, estimate an event-time specification:

$$\begin{aligned}
\log(K_{i,t} + 1) = & \alpha + \theta_1(\text{TREAT}_i \times \text{BEFORE}_{i,t}^{(-2,-1)}) + \theta_2(\text{TREAT}_i \times \text{CURRENT}_{i,t}) \\
& + \theta_3(\text{TREAT}_i \times \text{AFTER1}_{i,t}) + \theta_4(\text{TREAT}_i \times \text{AFTER2}_{i,t}) \\
& + \phi_1 \text{BEFORE}_{i,t}^{(-2,-1)} + \phi_2 \text{CURRENT}_{i,t} \\
& + \phi_3 \text{AFTER1}_{i,t} + \phi_4 \text{AFTER2}_{i,t} + \varepsilon_{i,t}.
\end{aligned} \tag{5}$$

Event-time dummies. Let year 0 be the privatization year.

- $\text{BEFORE}^{(-2,-1)}$: the two years prior to privatization ($t = -2, -1$),
- CURRENT : the privatization year ($t = 0$),
- AFTER1 : the first year after ($t = +1$),
- AFTER2+ : years $t \geq +2$.

Interpretation.

- θ_1 is a *pre-trend diagnostic*: it should be close to 0 if matching achieves parallel trends.
- θ_3, θ_4 capture dynamic divergence in innovation post-privatization for high-IPR vs low-IPR matched firms.

4.3 Model 3: IV for IPR protection (Table 8)

Motivation: even with DiD, local IPR protection may be correlated with other unobserved, innovation-relevant institutions (or with measurement error). The paper follows Ang, Cheng, and Wu (2014) and uses two historical province-level instruments:

- number of Christian colleges founded by missionaries in the province by 1920,
- indicator for whether the province had a British settlement (concession/leased territory) during the Qing dynasty.

These capture historical exposure to Western property-rights norms that plausibly affects current IPR institutions.

A schematic 2SLS system is:

$$\text{IPRScore}_{p,t} = \pi_0 + \pi_1 \text{ChristianCollege}_p + \pi_2 \text{BritishSettlement}_p + \Pi' W_{p,t} + u_{p,t}, \tag{6}$$

$$\log(K_{i,t} + 1) = \alpha + \beta_1 \text{POST}_{i,t} + \beta_2 \widehat{\text{IPRScore}}_{p(i),t-1} + \beta_3 (\text{POST}_{i,t} \times \widehat{\text{IPRScore}}_{p(i),t-1}) + \cdots + \varepsilon_{i,t}. \tag{7}$$

In practice, the interaction term is instrumented by interacting POST with the instruments.

5 Figure and table

5.1 Figure 1: geographic variation in IPR protection

What it is. A province-level map (aggregating prefecture-year IPR scores) showing strong regional heterogeneity.

How to read. Darker shading corresponds to higher average IPR scores (e.g., above 0.70). Coastal provinces are typically higher, but not perfectly so (e.g., some rich regions are not the highest).

Why it matters. The paper’s identification needs within-China cross-sectional variation in IPR enforcement quality; Figure 1 shows this is large.

5.2 Table 1: summary statistics of IPR protection by province

What it is. Province summaries of the prefecture-year IPR scores: mean, max, min, SD, and coastal indicator.

Key takeaways.

- There is meaningful *within-province* variation (max–min gaps) and cross-province variation in mean scores.
- The paper notes a positive correlation between IPR scores and coastal status, but the relationship is far from deterministic.

5.3 Figure 2: IPR score over time

What it is. Mean IPR score trends for prefectures above vs below the median each year.

Takeaway. Both high and low groups rise from 2002 to about 2009 and then soften slightly. This suggests broad improvement in perceived IPR enforcement, with persistent dispersion between high and low groups.

Table 1
Summary statistics on IPR protection index

Province	Coastal (Y/N)	Average IPR score	Within province variation		
			Max.	Min.	SD
1 Heilongjiang	N	0.58	0.81	0.42	0.13
2 Yunnan	N	0.59	0.84	0.33	0.17
3 Jilin	N	0.62	0.81	0.33	0.20
4 Inner Mongolia	N	0.63	0.80	0.38	0.30
5 Shanxi ^a	N	0.64	0.77	0.46	0.09
6 Chongqing	N	0.64	0.72	0.50	0.08
7 Hunan	N	0.64	0.82	0.48	0.13
8 Liaoning	Y	0.65	0.82	0.42	0.12
9 Henan	N	0.65	0.87	0.56	0.09
10 Hubei	N	0.66	0.90	0.42	0.16
11 Jiangxi	N	0.66	0.86	0.50	0.12
12 Shaanxi ^a	N	0.67	0.77	0.48	0.09
13 Sichuan	N	0.67	0.80	0.58	0.08
14 Hainan	Y	0.68	0.85	0.38	0.10
15 Guangdong	Y	0.70	0.96	0.25	0.17
16 Guangxi	Y	0.71	0.86	0.61	0.07
17 Beijing	N	0.71	1.00	0.55	0.14
18 Hebei	Y	0.72	0.87	0.58	0.08
19 Anhui	N	0.72	1.00	0.53	0.08
20 Fujian	Y	0.72	1.00	0.48	0.10
21 Tianjin	Y	0.74	0.87	0.58	0.09
22 Shandong	Y	0.75	0.95	0.50	0.09
23 Zhejiang	Y	0.76	1.00	0.50	0.09
24 Jiangsu	Y	0.76	1.00	0.42	0.12
25 Shanghai	Y	0.83	1.00	0.69	0.09

^aShaanxi is for 陕西省 in Chinese. Shanxi is for 山西省 in Chinese. These two Chinese provinces have the same phonetic spelling (Chinese pinyin) but with different meanings. We adopted a slightly different spelling to make the distinction.
This table reports summary statistics of IPR scores provided by the Chinese Academy of Social Sciences. The calculation of the IPR score is detailed in the text. The data are from 2002 to 2011 and are at the prefecture-year level. We calculate and report provincial-level statistics based on prefectures and years pertaining to each province. The provinces are listed in ascending order of the average IPR score.

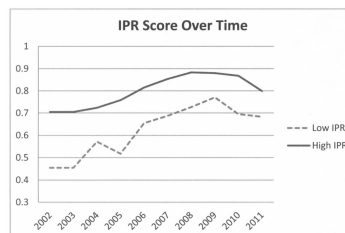


Figure 2
IPR score over time
This figure plots the mean IPR scores across prefectures over time. The IPR score data are from the Chinese Academy of Social Sciences, and its calculation is detailed in the text. Prefectures in the high (low) IPR group are those whose IPR scores are above (below) the median each year.

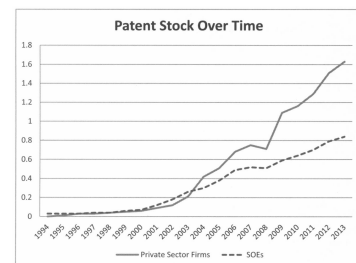


Figure 3
Innovation trend
This figure plots the patent stock over time for privately owned firms and state-owned enterprises (SOEs). Patent data are from the Chinese State Intellectual Property Office. Patent stock is calculated as in Equation (1) and scaled by million RMB in assets.

5.4 Figure 3: patent stock over time (private vs SOEs)

What it is. Patent stock per asset (scaled) for listed Chinese private firms vs SOEs from the mid-1990s to 2013.

Takeaway. Both groups increase, but private firms accelerate and overtake SOEs around the mid-2000s. This motivates studying ownership incentives, not just aggregate innovation growth.

5.5 Figure 4: policy-trend evidence (media mentions)

What it is. Frequency of official media mentions of “privatization” vs “innovation” in provincial government “Daily” newspapers, 2000–2013.

Takeaway. “Privatization” peaks before 2005 and declines; “innovation” rises strongly after 2011. This supports the claim that the privatization wave was not primarily driven by an innovation-focused policy agenda (helping the exogeneity narrative).

5.6 Table 2: number of privatizations by year and province coverage

What it is. Counts of privatizations (and number of provinces represented) by year.

Takeaway. The privatization wave is concentrated in the early-to-mid 2000s (peaking around 2004), then declines. This timing is useful for understanding the sample and for thinking about overlapping national reforms.

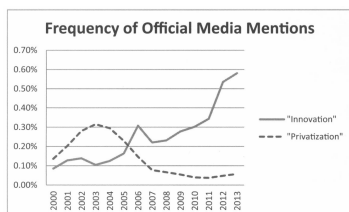


Figure 4
Policy trends: Privatization versus innovation
This figure plots the frequency of media mentions of the key words “Innovation” and “Privatization” in all the “Daily” newspapers, the official publications of provincial governments in China. The numbers are calculated by dividing the number of articles that contain each key word with the number of total articles in a given year, averaged across all provinces.

Year	Number of privatizations	Number of cities	Number of provinces
2000	175	103	27
2001	199	109	27
2002	110	70	25
2003	141	89	26
2004	242	116	29
2005	166	97	26
2006	232	133	29
2007	25	22	14
2008	10	10	7
2009	7	6	5
2010	6	5	5
2011	18	17	12

This table lists the basic information for our privatization sample. SOE privatizations are identified from the CSMAR database for listed firms and from the Industrial Enterprises Database from the National Statistical Bureau of China (NSBC) for unlisted firms. Section 1.3 discusses the details of the identification.

	Before	After	Diff	t-stat
Ln(asset)	17.831	18.036	0.206	-0.010
Intangible	0.009	0.010	0.001	0.093
ROA	0.030	0.044	0.014	0.003
Leverage	0.655	0.653	-0.002	-0.860

This table compares firm characteristics in the year before and the year after the privatization. Ln(asset) is the natural logarithm of total assets (measured in RMB). Intangible is the firm’s intangible assets divided by total assets. ROA is the net income divided by total assets. Leverage is total debt divided by total assets. All ratios are expressed in decimal points. t-statistics based on two-tailed tests are reported (none are statistically significant at conventional confidence thresholds).

	Before privatization	After privatization	difference	t-stat (diff)
Invention and utility patent stock	0.416	1.495	1.079	4.793***
Invention patent stock	0.147	0.461	0.314	3.338***
Utility patent stock	0.268	1.034	0.765	4.402***

This table compares firms’ patenting rates before and after privatization. The numbers tabulated are the average patent stock in the five years before and five years after the privatization. *, **, and *** represent statistical significance at the 10%, 5%, and 1% level, respectively.

5.7 Table 3: firm characteristics before vs after privatization (univariate)

What it is. Mean firm characteristics comparing pre vs post windows.

Takeaway. The differences in size, ROA, leverage, intangible ratio are small and not statistically large in this univariate comparison. This is *not* a definitive selection test, but it helps indicate that the innovation results are not mechanically driven by large shifts in these observables.

5.8 Table 4: innovation before vs after privatization (univariate)

What it is. Average patent stock in 5-year pre vs 5-year post windows.

Main message. Patent stock rises strongly after privatization for:

- invention+utility combined,
- invention-only,
- utility-only.

This establishes the baseline “privatization $\rightarrow$ more patenting” fact.

5.9 Table 5: high-IPR vs low-IPR DiD (univariate)

What it is. A clean DiD table:

$$(\text{After} - \text{Before})_{\text{High IPR}} - (\text{After} - \text{Before})_{\text{Low IPR}}.$$

Takeaway. The post-privatization increase in patent stock is much larger in high-IPR regions, for both invention and utility patents. This is the first direct evidence for the interaction hypothesis.

5.10 Table 6: baseline panel regressions (Eq. (4))

What it is. Panel regressions with controls and fixed effects.

Key result. The interaction coefficient β_3 on $\text{POST} \times \text{IPRScore}$ is positive and statistically significant across specifications, implying:

$$\text{Stronger IPR} \Rightarrow \text{larger post-privatization innovation gain.}$$

Economic magnitude (as emphasized in the text). Using the estimates in column (2), at a mean IPR of about 0.65, the net post-privatization effect is near zero, but it increases markedly with a one-standard-deviation increase in IPR, consistent with the idea that IPR protection is what enables the incentive effects of privatization to translate into innovation. **Cross-patent-type robustness.** Columns (4)–(5) show the interaction logic holds for invention and utility separately.

Within-firm robustness. Column (6) includes firm fixed effects and still finds a positive interaction, alleviating concerns that time-invariant firm heterogeneity drives the results.

Table 5
Interactive effect of ownership and IPR protection: Difference-in-differences

A. Invention and utility patent stock				
	High IPR protection	Low IPR protection	Diff (high-low)	Diff-in-diff
Before	0.579	0.177	0.402***	
After	2.474	0.528	1.946***	
After - Before	1.896***	0.351***		1.545***
B. Invention patent stock				
	High IPR protection	Low IPR protection	Diff (high-low)	Diff-in-diff
Before	0.219	0.047	0.172**	
After	0.801	0.137	0.664***	
After - Before	0.582***	0.089**		0.492***
C. Utility patent stock				
	High IPR protection	Low IPR protection	Diff (high-low)	Diff-in-diff
Before	0.36	0.13	0.230***	
After	1.674	0.392	1.282***	
After - Before	1.314***	0.262***		1.052***

This table reports the results of a difference-in-differences analysis in which we compare the increase in patent stock before and after privatizations in prefectures with high and low IPR protection. The numbers tabulated are the average patent stock in the five years before and five years after privatization. High (low) IPR protection is based on whether a prefecture's IPR score is above (below) the median score in the year of a privatization event. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Table 6
Ownership type, IPR protection, and innovation: Regression analysis

DEP VAR = log of	(1)	(2)	(3)	(4)	(5)	(6)
	Invention and utility patent stock + 1	Invention and utility patent stock + 1	Invention and utility patent stock + 1	Invention patent stock + 1	Utility patent stock + 1	Invention and utility patent stock + 1
POST	0.025** (0.049)	-0.138** (0.037)	-0.027 (0.192)	-0.088** (0.041)	-0.091 (0.107)	-0.192** (0.034)
IPR	0.244** (0.014)	-0.116 (0.236)		-0.058 (0.385)	-0.131 (0.117)	-0.267* (0.099)
POST × IPR		0.225** (0.036)		0.138* (0.057)	0.158* (0.083)	0.327** (0.024)
IPR dummy			-0.038 (0.133)			
POST × IPR dummy			0.071** (0.016)			
Log(asset)	0.145*** (0.014)	0.137*** (0.000)	0.137*** (0.001)	0.074*** (0.000)	0.101*** (0.000)	0.067** (0.014)
Intangible	-0.256 (0.302)	-0.497** (0.045)	-0.499** (0.044)	0.07 (0.708)	-0.811*** (0.000)	0.372 (0.276)
ROA	0.318*** (0.009)	0.316*** (0.000)	0.319*** (0.001)	0.092* (0.067)	0.276*** (0.000)	-0.01 (0.886)
Leverage	-0.040** (0.040)	-0.048** (0.009)	-0.049*** (0.008)	-0.007 (0.588)	-0.043** (0.008)	0.044 (0.203)
Age	-0.030* (0.052)	-0.039*** (0.000)	-0.039*** (0.001)	-0.041*** (0.000)	-0.014* (0.086)	0.215*** (0.007)
GDP growth	0.204 (0.384)	-0.584** (0.012)	-0.580** (0.013)	-0.571*** (0.001)	-0.226 (0.216)	-0.075 (0.593)
University density	3.522** (0.013)	51.003*** (0.000)	51.576*** (0.001)	37.656*** (0.000)	33.019*** (0.000)	21.763* (0.054)
Gov't subsidy	59.492 (0.200)	74.302 (0.106)	74.209 (0.107)	50.783* (0.065)	72.532* (0.071)	8.484 (0.596)
Constant	-3.149*** (0.000)	-2.706*** (0.000)	-2.769*** (0.001)	-1.679*** (0.000)	-1.803*** (0.000)	-1.785** (0.012)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	No
Province fixed effects	Yes	Yes	Yes	Yes	Yes	No
Firm fixed effects	No	No	No	No	No	Yes
Observations	5,854	5,854	5,854	5,854	5,854	5,854
R-squared	0.235	0.270	0.27	0.212	0.242	0.865

This table examines the relation between ownership type, IPR protection, and innovation in a panel regression. The sample consists of firm-year observations of firms that have undergone a privatization between 2000 and 2011. For each privatization event, we use five years of data before and five years of data after the event as observations. The key regression we estimate is

$$\log(\text{Patent_Stock}_{i,t} + 1) = a + b_1 \text{POST}_{i,t} + b_2 \text{IPR}_{i,t-1} + b_3 \text{POST}_{i,t} \times \text{IPR}_{i,t-1} + \text{Controls}_{i,t-1} + \varepsilon_{it}$$

5.11 Table 7 and Figure 5: matched-sample evidence and dynamic effects

Panel A (balance). Treatment and control firms are similar on pre-privatization characteristics and, crucially, on pre-privatization patent-growth trends.

Panel B (DiD in matched sample). The high-IPR treatment group exhibits a larger post-privatization patent-stock increase than the matched low-IPR group.

Panel C (event-time regression). The pre-trend interaction is close to zero, while post interactions—especially $\text{TREAT} \times \text{AFTER2}+$ —are positive and significant. This says:

- *parallel trends* hold in the immediate pre-period (a key DiD diagnostic),
- the treatment effect accumulates over time (consistent with lags in innovation-to-patent realization).

Table 7
Matched sample analysis

A. Differences in pre-privatization characteristics

	Treatment	Control	Difference	t-statistic
Patent stock growth -5 to -1	0.372	0.274	0.098	0.784
Log(asset) -1	17.462	17.302	0.159	1.466
Intangible -1	0.003	0.002	0.001	0.867
ROA -1	0.019	0.021	-0.002	-0.330
Leverage -1	0.067	0.066	0.001	0.034
Age -1	2.746	2.766	-0.020	-0.320

B. Difference-in-differences test

	Treatment (high IPR)	Control (low IPR)	Diff (treatment-control)	Diff-in-diff
Before	0.207	0.154	0.052	
After	1.198	0.537	0.661*	
After-before	0.991***	0.383***		0.608**

(1) Invention and utility patent stock

	Treatment (high IPR)	Control (low IPR)	Diff (treatment-control)	Diff-in-diff
Before	0.207	0.154	0.052	
After	1.198	0.537	0.661*	
After-before	0.991***	0.383***		0.608**

(2) Invention patent stock

	Treatment (high IPR)	Control (low IPR)	Diff (treatment-control)	Diff-in-diff
Before	0.041	0.039	0.002	
After	0.295	0.11	0.183**	
After-before	0.255***	0.071*		0.181***

(3) Utility patent stock

	Treatment (high IPR)	Control (low IPR)	Diff (treatment-control)	Diff-in-diff
Before	0.166	0.116	0.05	
After	0.905	0.427	0.478*	
After-before	0.739***	0.312***		0.428**

Table 7
Continued

C. Regression results

Dependent variable	Ln (Invention and utility patent stock + 1)	Ln (Invention patent stock + 1)	Ln (Utility patent stock + 1)
Treat × Before ⁻² & -1	0.0052 (0.820)	-0.007 (0.520)	0.0144 (0.470)
Treat × Current	-0.0088 (0.790)	-0.0025 (0.880)	0.0176 (0.560)
Treat × After ¹	0.0431 (0.250)	0.011 (0.210)	0.0421 (0.210)
Treat × After ²	0.0783*** (0.000)	0.0461*** (0.000)	0.0627*** (0.010)
Before ⁻² & -1	0.0196*** (0.000)	0.0195** (0.000)	0.0322** (0.010)
Current	0.0002*** (0.000)	0.0002*** (0.000)	0.0002*** (0.000)
After ¹	0.0002*** (0.000)	0.0002*** (0.000)	0.0002*** (0.000)
After ²	0.0002*** (0.000)	0.0002*** (0.000)	0.0002*** (0.000)
Abatch ²	0.1356*** (0.000)	0.0484*** (0.000)	0.1095*** (0.000)
Treat	0.0131 (0.260)	0.0056 (0.410)	0.0085 (0.410)
Constant	0.0091*** (0.000)	0.0091*** (0.000)	0.0096*** (0.000)
Observations	8,081	8,081	8,081
R-squared	0.026	0.018	0.021

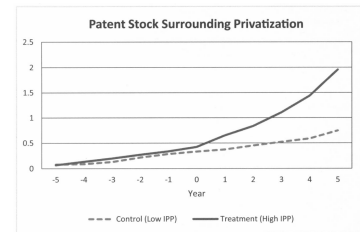


Figure 5
Privatizations and innovation: Matched sample analysis
This figure shows the mean patent stock for treatment and control firms, from five years before the privatization to five years after the privatization. The privatization year is year 0. The sample comprises 380 treatment firms (firms located in high IPR protection regions) and 380 unique control firms located in low IPR protection regions) based on the propensity score matching procedures described in Section 2.2.

5.12 Table 8: IV results

First stage. Historical instruments strongly predict IPR score (large first-stage F-statistic).

Second stage. The instrumented interaction $POST \times \widehat{IPR}Score$ remains positive and significant. This supports a causal interpretation: IPR institutions, not merely correlated regional development, are shaping how privatization affects innovation.

Table 8
Instrumental variables regression

A. First-stage results	
DEPVAR =	IPR
Christian college	0.017*** (0.000)
British settlement	0.031*** (0.000)
Gov't subsidy	-8.891 (0.266)
University density	3.044*** (0.000)
GDP growth	0.681*** (0.000)
Observations	5,854
R-squared	0.226
Partial-F test for IVs	143.84 (p=0.000)

B. Second-stage results

	(1)	(2)	(3)
DEP VAR = log of	Invention and utility patent stock + 1	Invention patent stock + 1	Utility patent stock + 1
POST	-0.340** (0.049)	-0.215* (0.064)	-0.206 (0.162)
IPR (IV)	0.448 (0.156)	0.133 (0.529)	0.181 (0.502)
POST $\times$ IPR (IV)	0.523** (0.048)	0.344* (0.052)	0.327* (0.097)
Controls	Yes	Yes	Yes
Observations	5,854	5,854	5,854
R-squared	0.234	0.18	0.209
Sargan-Hansen's J statistic (overidentifying restrictions)	1.715 (p=0.424)	4.251 (p=0.119)	0.717 (p=0.699)

This table reports instrumental variables (IV) regression results. In the first stage, we predict IPR scores using two instrumental variables: Christian college (the number of colleges founded by Christian missionaries in a province before 1920) and British settlement (a dummy variable that equals one if the province had a British concession in the Qing Dynasty and zero otherwise). In the second stage, the unit of observation is (as in Table 6) annual observations of each privatized firm for the five years before and the five years after the privatization. The dependent and independent variables are identical to those in regressions (1) through (3) of Table 6, with the exception that we employ the instrumented value for the IPR protection measure. p -values are reported in parentheses. *, **, and *** indicates statistical significance at the 10%, 5%, and 1% level, respectively.

Table 9
Patent quality

A. Entire sample: Patents published between 1990 and 2014

	Private	SOE	t -stat (diff)	
Citations received per patent	0.76	0.47	2.74	***
IPC groups per patent	1.81	1.66	3.15	***
% patents active in the U.S.	6.1%	4.2%	2.40	**
% patents active in Japan	1.6%	1.9%	-0.90	
% patents active in EPO	0.6%	0.9%	-1.21	
% patents active in WIPO	1.4%	0.5%	3.15	***

B. Period 1: Patents published between 1990 and 2005

Citations received per patent	2.89	1.17	4.06	***
IPC groups per patent	1.89	1.77	1.04	
% patents active in U.S.	17.6%	9.9%	2.81	***
% patents active in Japan	5.6%	3.8%	1.20	
% patents active in EPO	0.3%	0.5%	-0.52	
% patents active in WIPO	0.0%	0.0%	.	

C. Period 2: Patents published between 2006 and 2014

Citations received per patent	0.30	0.19	2.17	**
IPC groups per patent	1.80	1.61	3.31	***
% patents active in the U.S.	3.6%	2.0%	2.76	***
% patents active in Japan	0.7%	1.2%	-1.84	**
% patents active in EPO	0.6%	1.0%	-1.31	
% patents active in WIPO	1.8%	0.7%	2.72	***

The table presents global patent application and citation data for the 331 publicly listed firms in China with the most Chinese patents. Of these 331 firms, 162 are SOEs (49%) and 169 (51%) private enterprises. "Citations received per patent" is the number of citations received worldwide per patent family owned by each firm as of year-end 2014. "IPC groups per patent" is the mean number of IPC groups that each patent family has been assigned to (a proxy for patent scope). "% patents active in U.S./Japan/European Patent Offices (EPO)" is the fraction of a firm's patent families awarded through that date that have patents that are pending and/or are granted in the respective patent offices as of year-end 2014. "% patent active in WIPO" is the share of the firm's patents active under the Patent Cooperation Treaty as of year-end 2014. We examine these patterns using all patent families (in panel A), those first published between the beginning of 1990 and the end of 2005 (panel B), and those first published between 2006 and 2014 (panel C). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level using a one-tailed test, respectively.

5.13 Table 9: patent quality (private vs SOE patents)

What it is. A comparison of global patent outcomes for 331 Chinese listed firms with many patents: citations per patent, IPC breadth, and international coverage (US/Japan/EPO/WIPO activity).

Main message. Private firms' patents are on average more cited and more internationally active (especially in the US and via WIPO), suggesting the private-sector innovation advantage is not purely "trivial patent inflation".

6 Short summary

1. Ownership incentives matter for innovation, but institutions determine whether those incentives can pay off.
2. Privatization is an ownership shock; local IPR is an appropriation friction.
3. Empirically: $\text{POST} \times \text{IPR}$ is the key interaction, and it is robust (controls, FE, matching, IV).

7 Conclusion

7.1 Coherent summary: what the paper ultimately establishes

This paper identifies a **complementarity between ownership incentives and institutional quality** in generating innovation. Using a difference-in-differences design around China’s large SOE privatization wave and cross-prefecture variation in intellectual property rights (IPR) protection, the authors conclude:

1. **Privatization raises innovation.** Innovation (measured by patent stock in a five-year window) rises substantially after SOEs are privatized. In the paper’s own summary, mean patent stocks are about **200%–300% larger** in the five years after privatization than in the five years before.
2. **IPR protection is a powerful moderator of the privatization–innovation effect.** The post-privatization increase in innovation is *much* larger in locations with stronger IPR protection. A transparent univariate difference-in-differences illustrates the magnitude (Table 5, Panel A):

$$\Delta \text{PatentStock}_{\text{High IPR}} \approx 2.474 - 0.579 = 1.896 \quad \text{vs.} \quad \Delta \text{PatentStock}_{\text{Low IPR}} \approx 0.528 - 0.177 = 0.351,$$

so the implied difference-in-differences is approximately

$$(1.896 - 0.351) \approx 1.545.$$

A similar pattern appears for invention patents (Table 5, Panel B), with a large positive diff-in-diff.

3. **The interaction survives in multivariate panel regressions.** Estimating the core specification

$$\log(1 + \text{PatentStock}_{i,t}) = a + b_1 \text{POST}_{i,t} + b_2 \text{IPR}_{\ell,t-1} + b_3 (\text{POST}_{i,t} \times \text{IPR}_{\ell,t-1}) + \Gamma' \text{Controls}_{i,t-1} + \text{FE} + \varepsilon_{i,t}, \quad (8)$$

the key complementarity term b_3 is positive and statistically significant (Table 6). For example, in the baseline interaction column (Table 6, col. (2)),

$$\hat{b}_3 \approx 0.225 \quad (p \approx 0.036),$$

and the interaction remains positive for invention and utility patents and in a specification with firm fixed effects.

4. **Endogeneity concerns about IPR are addressed with an IV strategy.** The authors instrument IPR using historically predetermined proxies for Western legal/educational influence (Christian colleges founded by missionaries before 1920; British settlement/concession exposure). The first stage is very strong (partial F-statistic ≈ 143.84 ; Table 8). In the second stage, the interaction between post-privatization status and instrumented IPR remains positive and significant (e.g., Table 8 Panel B shows $\hat{b}_3^{IV} \approx 0.523$ for total patents, with $p \approx 0.048$), reinforcing a causal interpretation that **stronger IPR protection amplifies the innovation response to privatization**.
5. **The increase is not merely “window dressing”: private-sector patents are higher quality.** Patent-quality evidence (Table 9) shows private firms’ patents are not lower quality than SOE patents. In the full 1990–2014 sample:

Citations per patent: 0.76 (private) > 0.47 (SOE), IPC groups per patent: 1.81 > 1.66,

and private firms’ patent families are more likely to be active internationally (e.g., U.S. and WIPO activity shares are higher).

7.2 Economic intuition: why these results make sense

The mechanism is a **two-part incentive system** in which ownership reform changes the payoff to innovation, and IPR institutions determine whether those payoffs can be appropriated:

- **Privatization increases residual-claim incentives.** Moving from state control toward private ownership strengthens managerial and shareholder incentives to pursue value-enhancing, risky, long-horizon investments such as R&D and patenting.
- **IPR protection determines appropriability.** Innovation requires upfront cost and uncertain returns. When IPR enforcement is weak, imitation and expropriation risk is high, so the expected private return to innovation is low even if the firm becomes privately owned. When IPR enforcement is strong, innovators can appropriate more of the upside, so incentives to innovate become much sharper.
- **Complementarity (the key conceptual contribution).** Ownership incentives and legal institutions are *complements*, not substitutes: privatization alone does not guarantee a large

innovation response unless the local institutional environment (IPR protection) supports appropriation; conversely, strong IPR protection matters most when firms have private-sector incentives.

7.3 Broader implications

- **Resolving the “China puzzle” in law and finance:** within a single country, stronger institutions (here, IPR protection) are linked to stronger innovative outcomes, consistent with a Schumpeterian view that institutions matter for growth-related activities.
- **Policy implication:** an innovation-driven growth model in China likely requires both (i) a dynamic private sector and (ii) credible legal protection of intellectual property; strengthening IPR institutions is especially consequential for private-firm innovation.

7.4 Takeaway

SOE privatization raises innovation, but the innovation response is concentrated in places with strong IPR protection, implying that private ownership incentives and IPR institutions are complementary determinants of innovative activity.